

1.B Preference Relations

In the preference-based approach, the objectives of the decision maker are summarized in a *preference relation*, which we denote by $\succsim$. Technically, $\succsim$ is a binary relation on the set of alternatives X , allowing the comparison of pairs of alternatives $x, y \in X$. We read $x \succsim y$ as “ x is at least as good as y .” From $\succsim$, we can derive two other important relations on X :

- (i) The *strict preference* relation, $\succ$, defined by

$$x \succ y \Leftrightarrow x \succsim y \text{ but not } y \succsim x$$

and read “ x is preferred to y .”¹

- (ii) The *indifference* relation, $\sim$, defined by

$$x \sim y \Leftrightarrow x \succsim y \text{ and } y \succsim x$$

and read “ x is indifferent to y .”